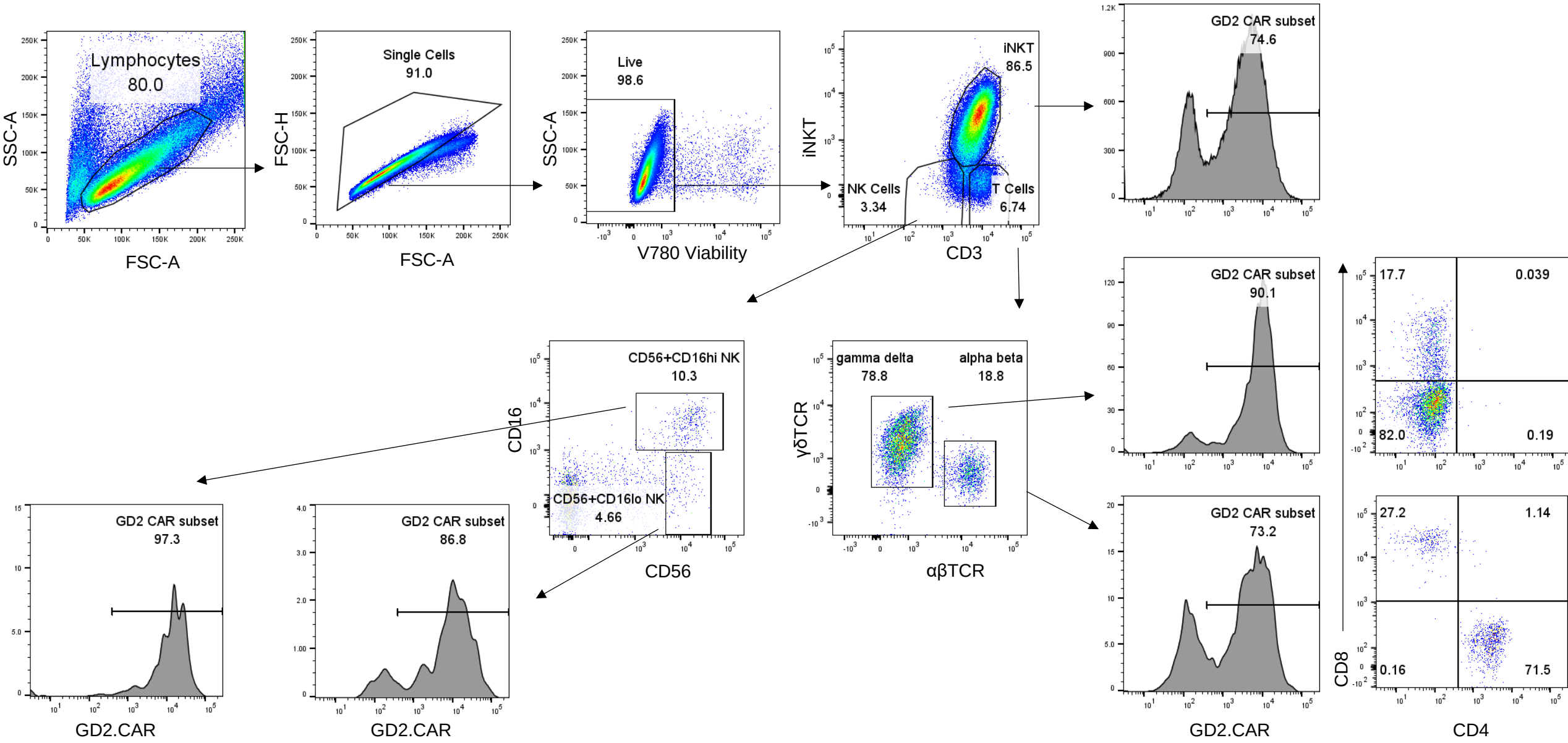
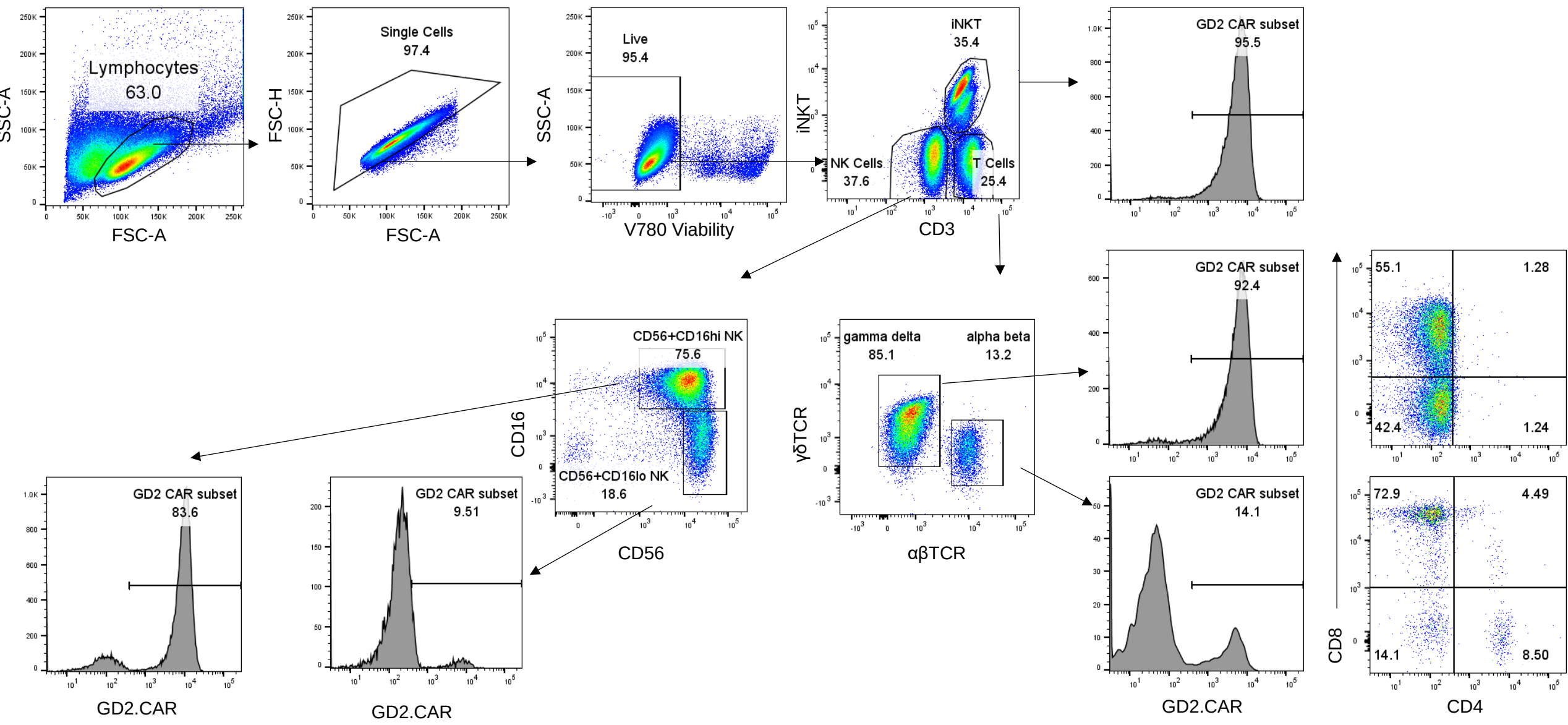


Pre-infusion product

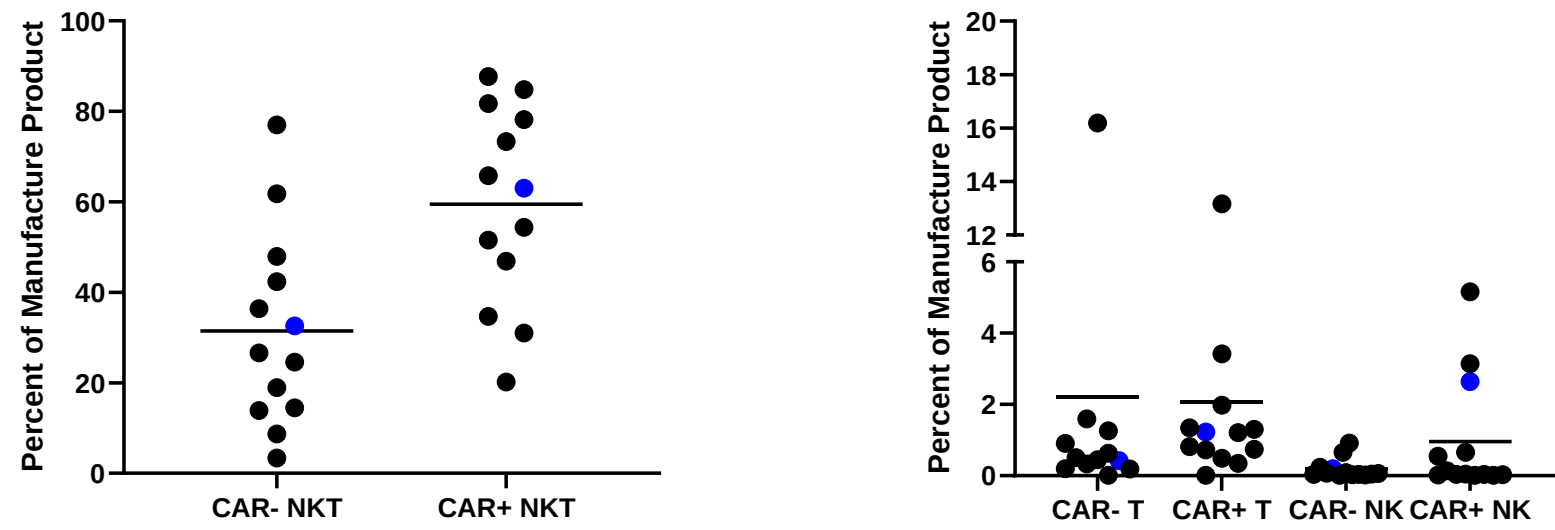


Day 14 after infusion

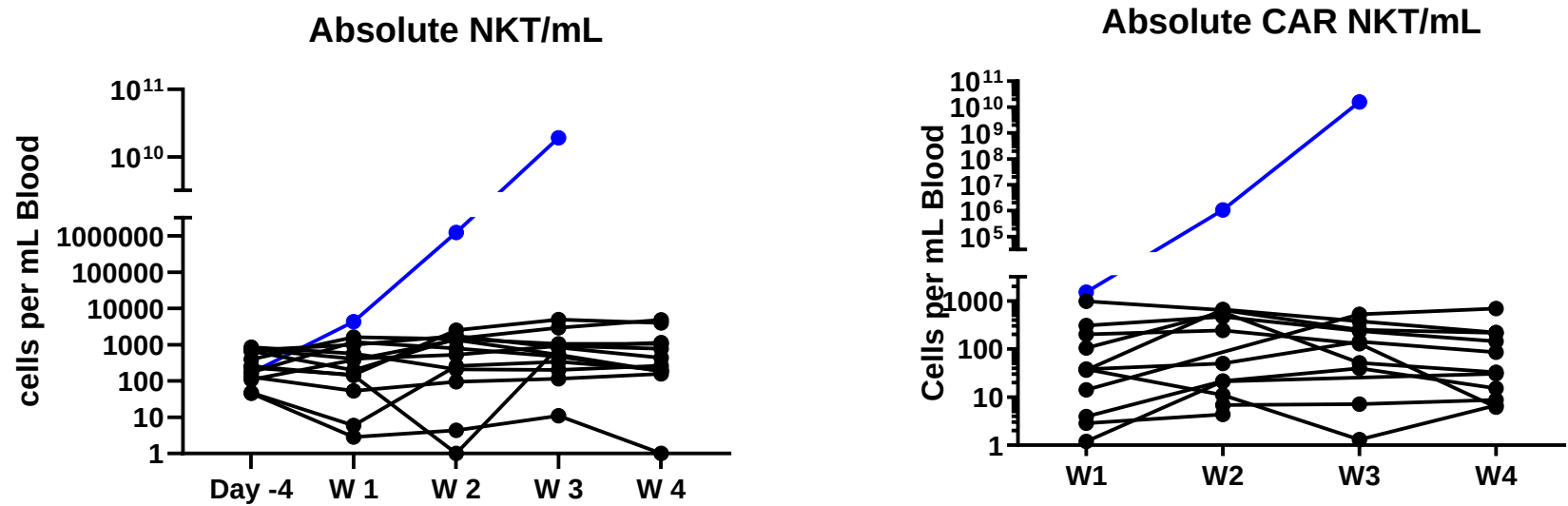


Supplemental Figure 1. NK and γ/δ T cells expanded to a greater extent than NKT and α/β T cells in patient P5-1. FACS cell subset analysis of patient P5-1 infusion product (top) and PBMCs isolated from patient P5-1 14 days post-infusion (bottom). Gating strategy shows relative proportions of NKT, NK, α/β T, and γ/δ T cells with CAR expression in each population.

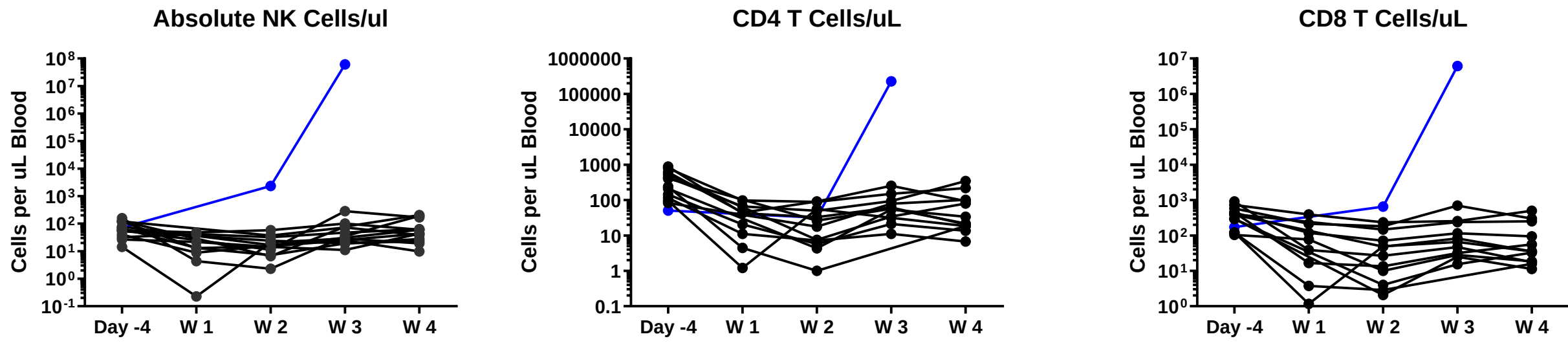
A



B

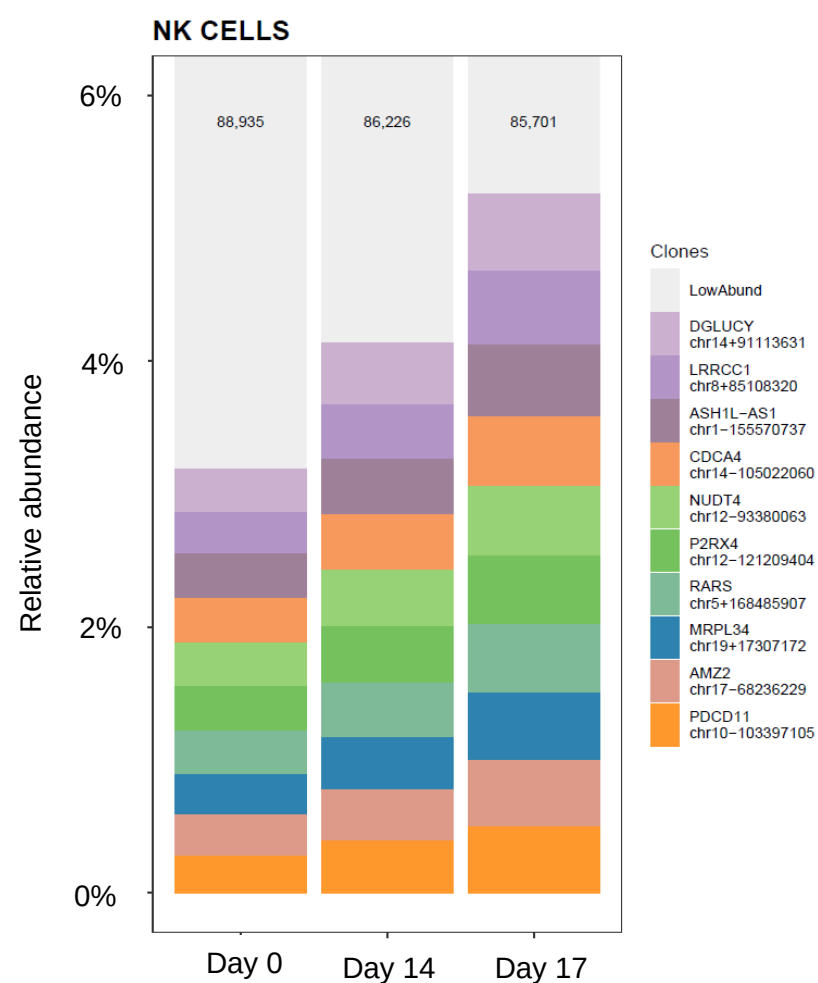
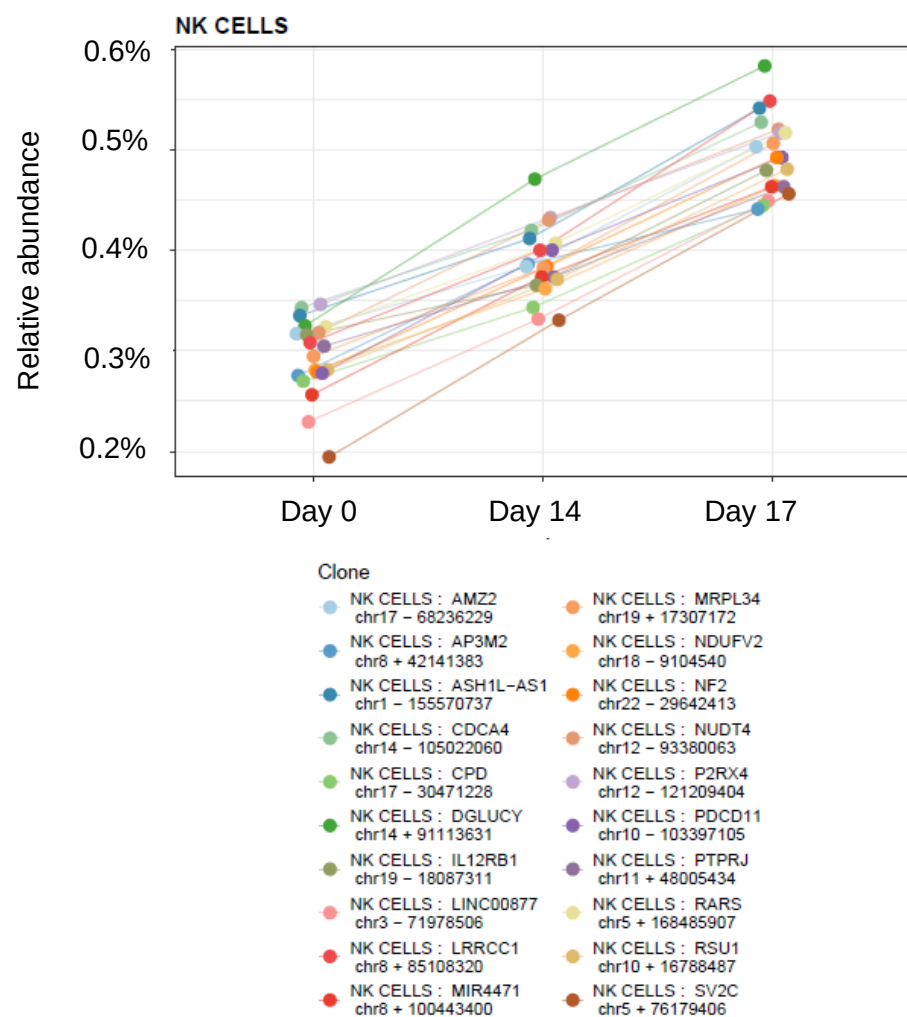


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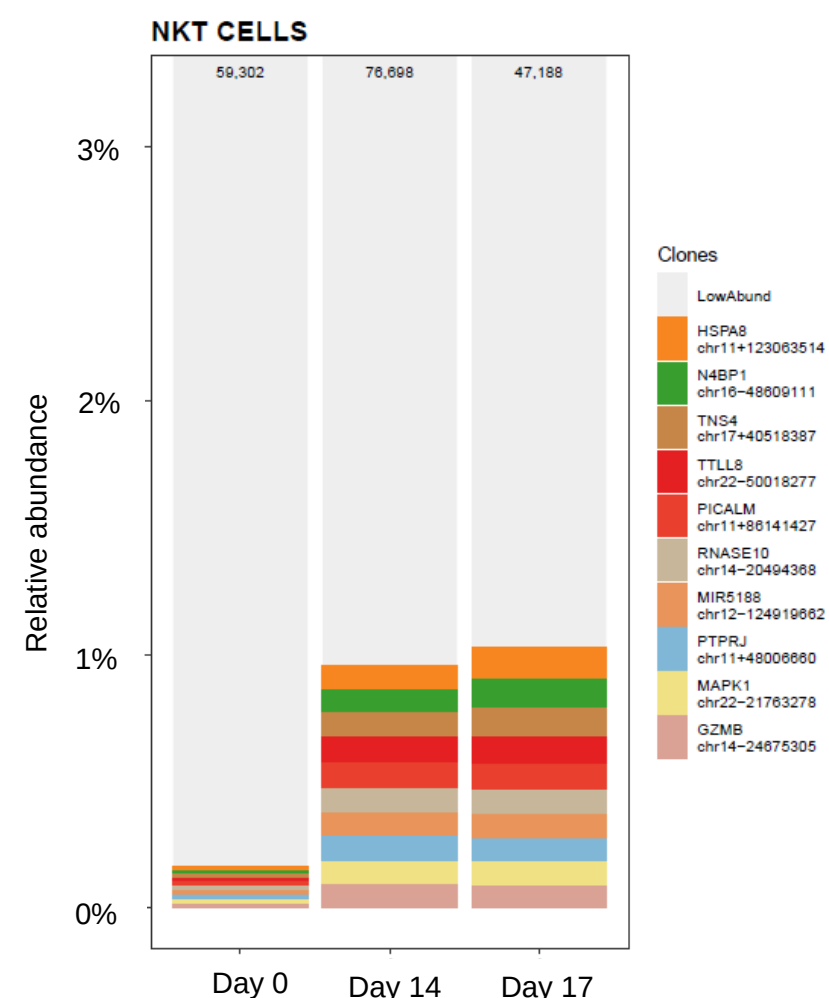
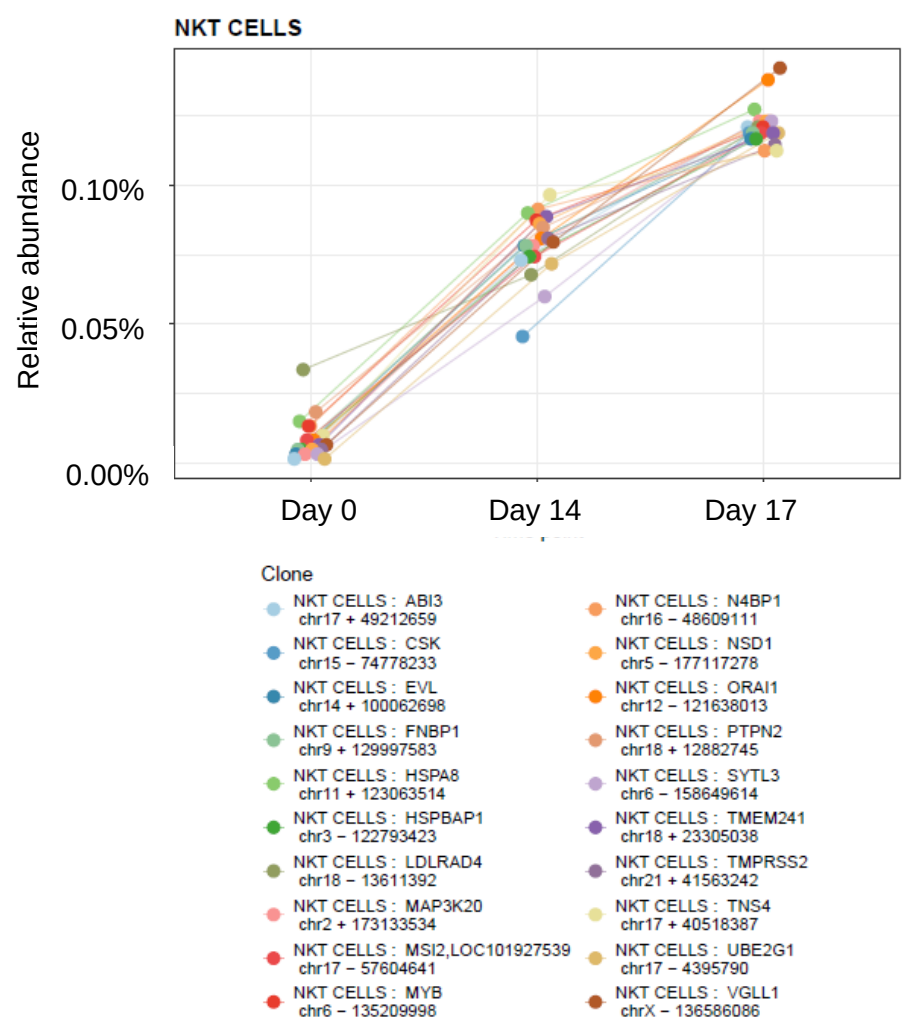


Supplemental Figure 2. Analysis of CAR+/- NKT and NK, CD4, & CD8 T cell population numbers in pre-infusion products and peripheral blood collected one to four weeks after infusion for P5-1 versus 12 other patients. A) CAR+ and CAR- NKT percentage in pre-infusion products (left); CAR+ and CAR- T and NK cell percentages in pre-infusion products (right). Blue dot represents P5-1. B) NKT absolute numbers (left) and CAR+ NKTs (right) in peripheral blood over four-week monitoring period. Blue lines represent P5-1. C) Absolute numbers of NK (left), CD4 T (middle), and CD8 T (right) cells in peripheral blood over four-week monitoring period. Blue lines represent P5-1.

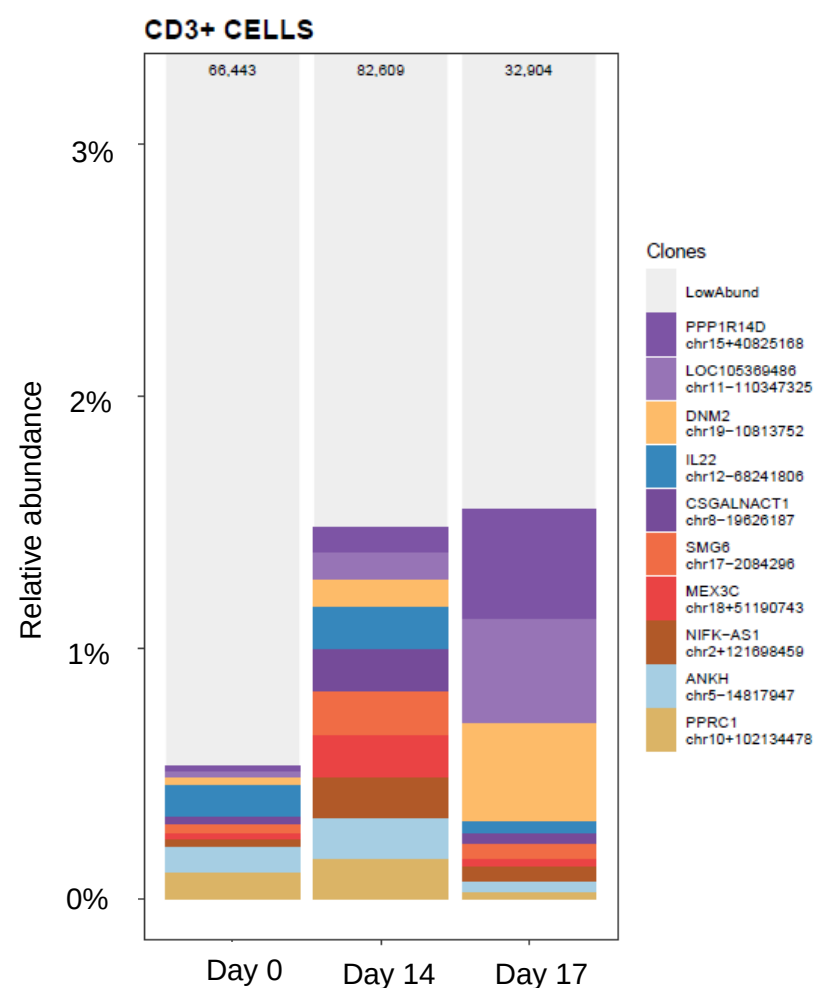
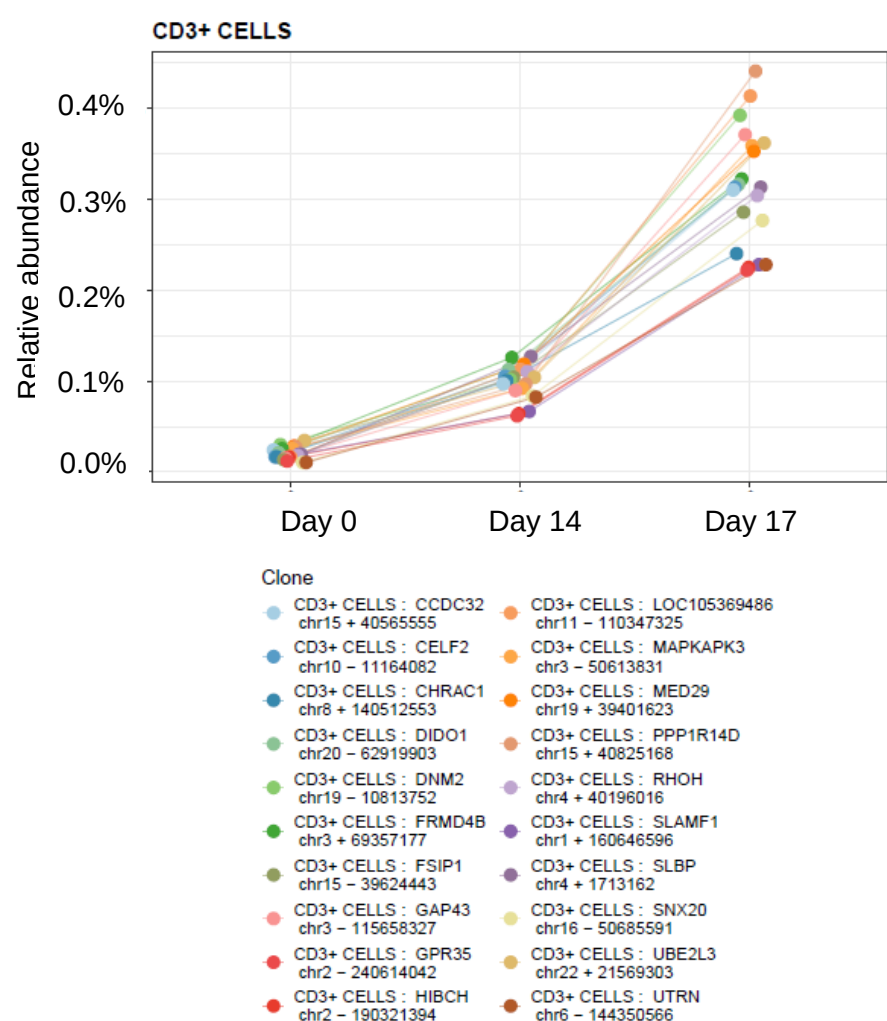
A



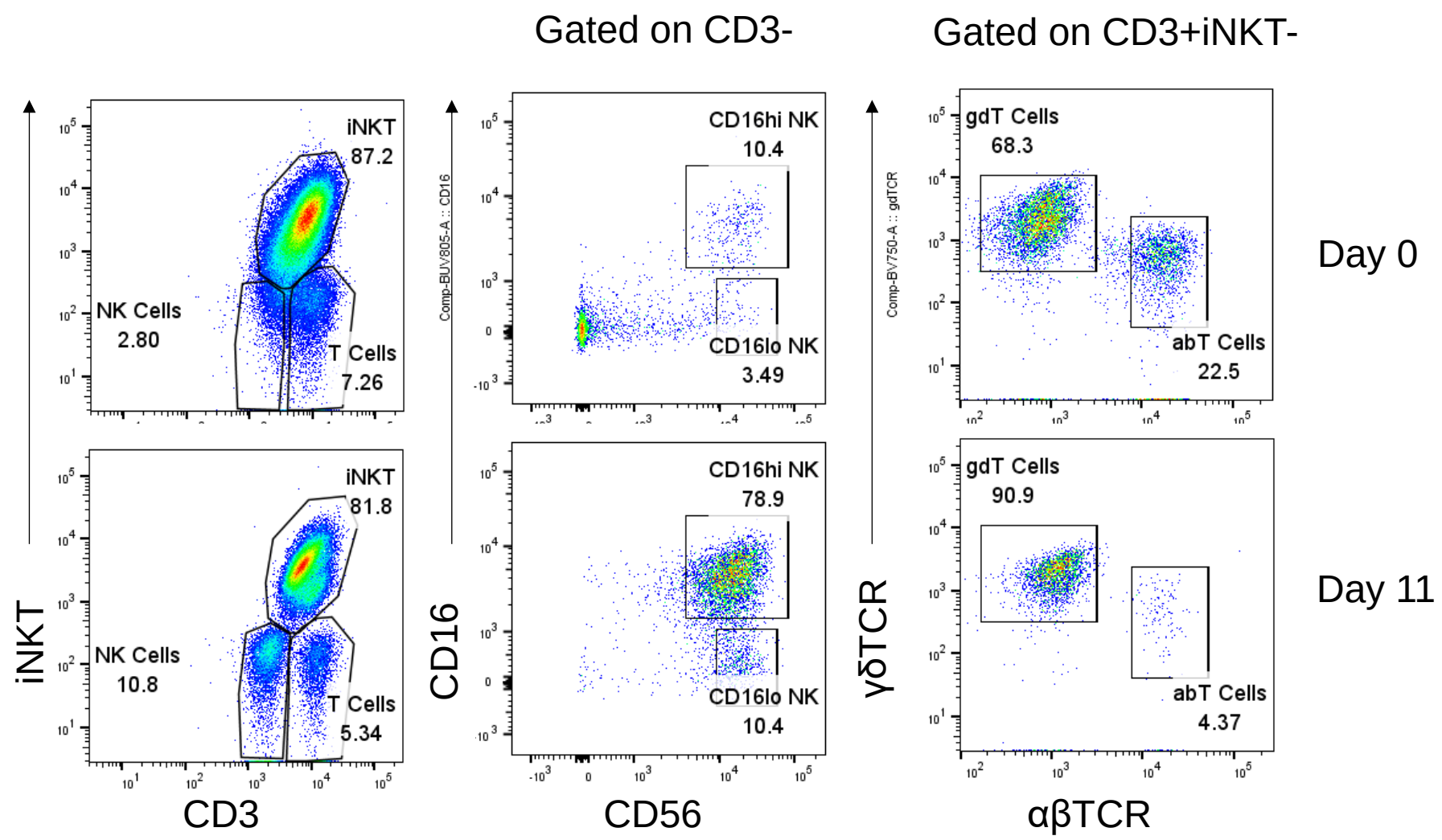
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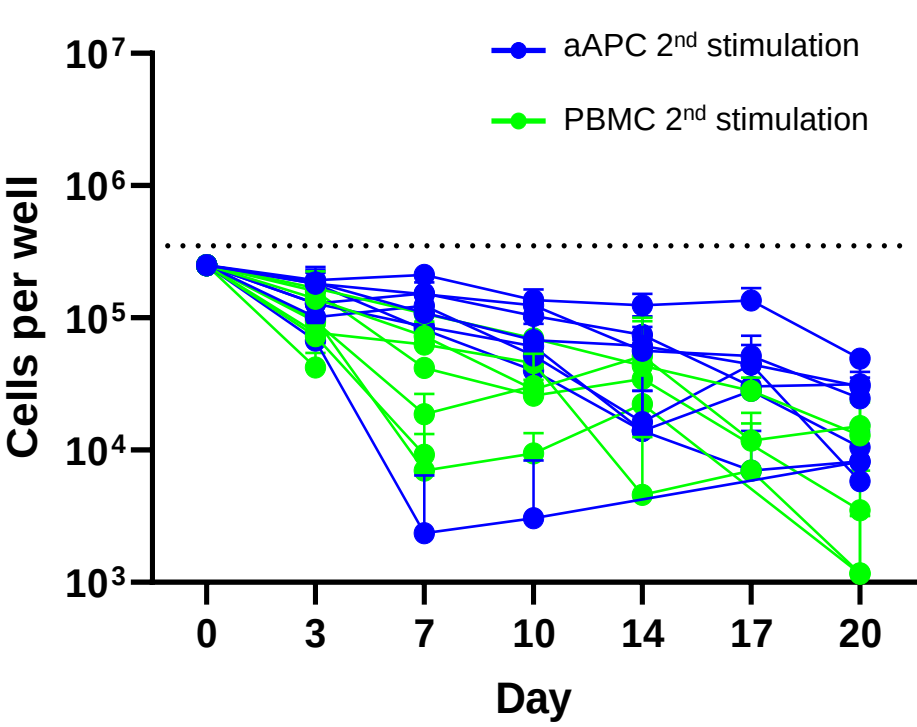
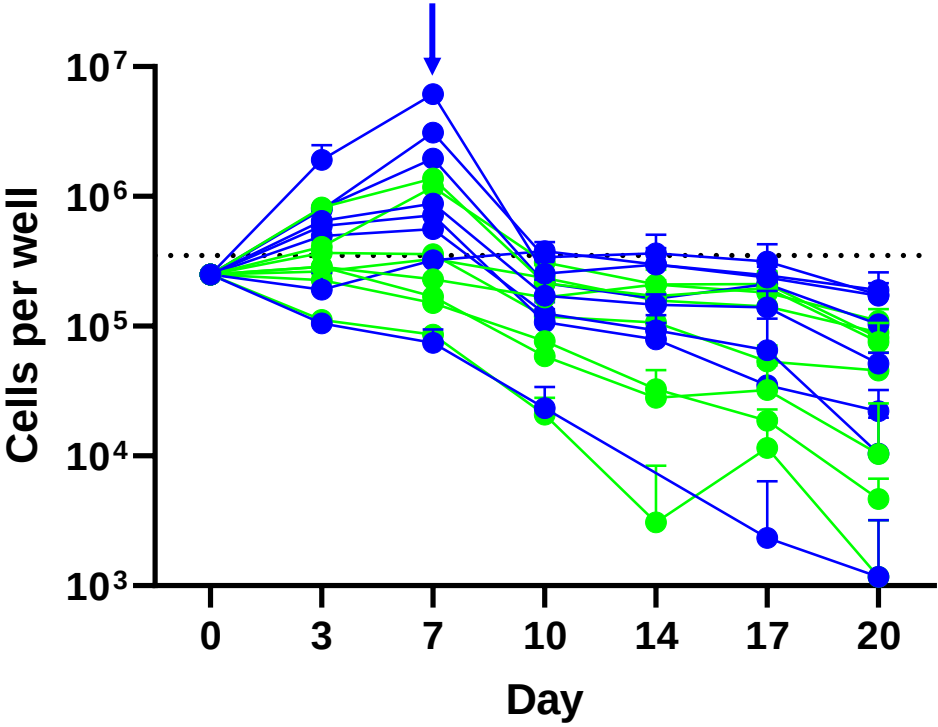


Supplemental Figure 3. Analysis of integration site distributions and relative clonal abundance. A) Longitudinal sample showing relative abundances of the top 20 most abundant clones of sorted NK cells (left) and stacked bar plot showing relative abundances of the top 10 major cell clones marked by integration sites from CAR-NK cells sorted from the infusion product, day 14 and day 17 post-infusion samples (right). B) Longitudinal sample showing relative abundances of the top 20 most abundant clones of sorted NKT cells (left) and stacked bar plot showing relative abundances of the top 10 major cell clones marked by integration sites from sorted CAR-NKT cells in the infusion product, day 14 sorted and day 17 post-infusion samples (right). C) Longitudinal sample showing relative abundances of the top 20 most abundant clones of sorted T cells (left) and stacked bar plot showing relative abundances of the top 10 major cell clones marked by integration sites from sorted CAR-T cells in the infusion product, day 14 and day 17 post-infusion samples (right).

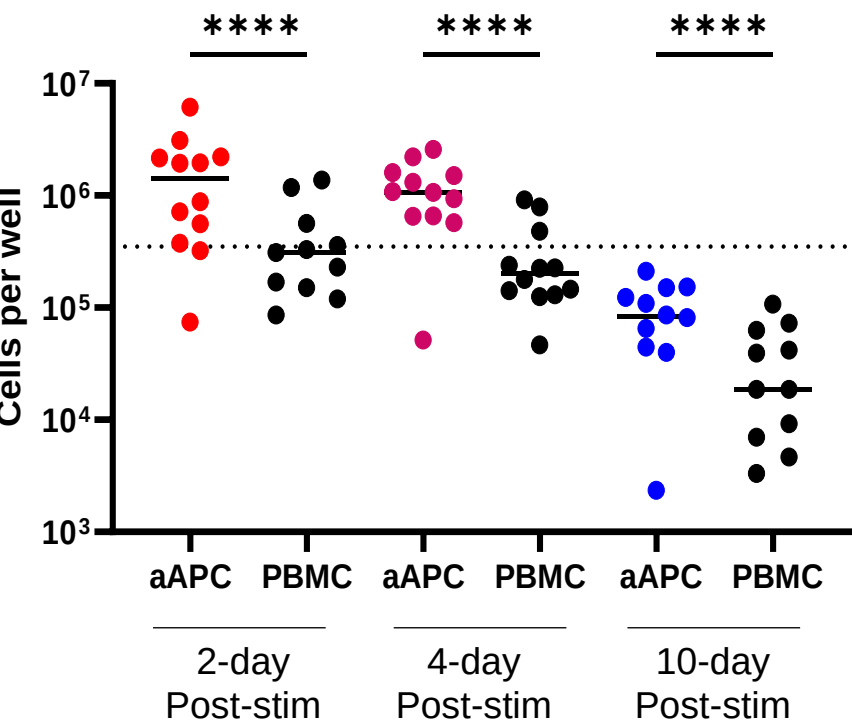


Supplemental Figure 4. NK and γ/δ T cells in patient P5-1 pre-infusion product expanded to a greater extent than NKT and α/β T cells during *ex vivo* growth assay. Relative frequencies of NKT, NK (CD16dim and CD16bright), α/β , and γ/δ T cell subsets from patient P5-1 pre-infusion product at day 0 and day 11 of *ex vivo* growth assay without exogenous cytokines.

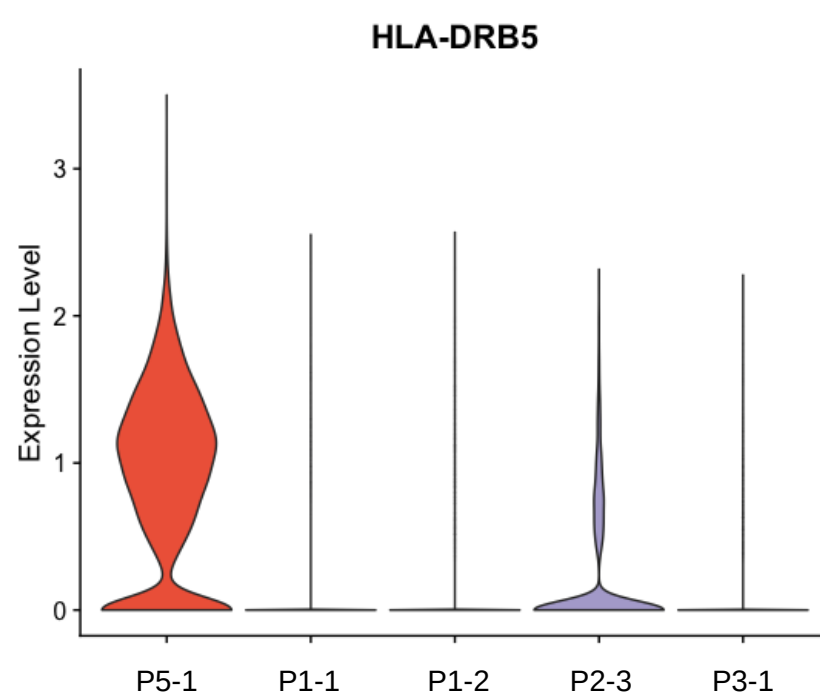
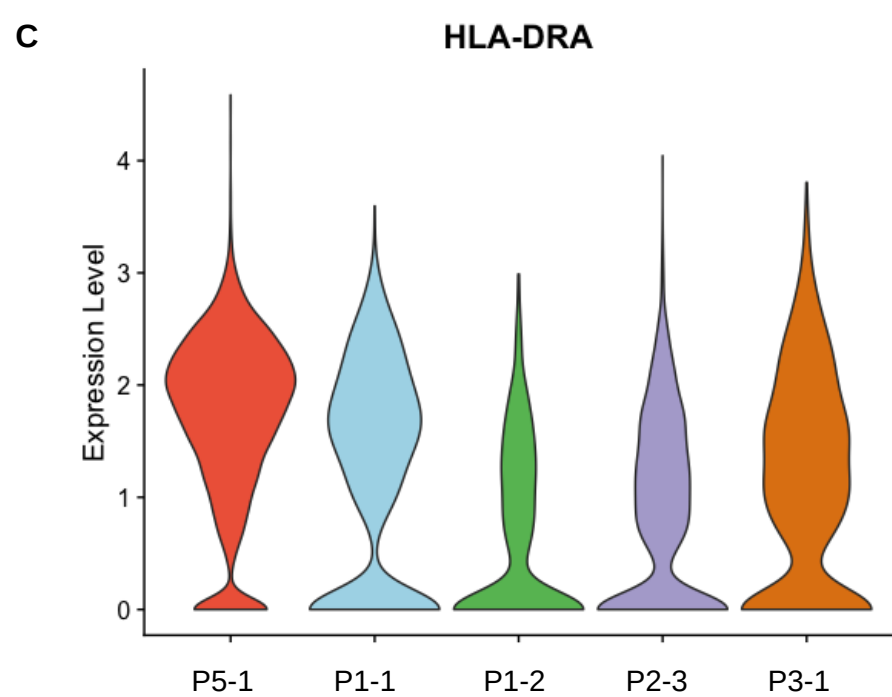
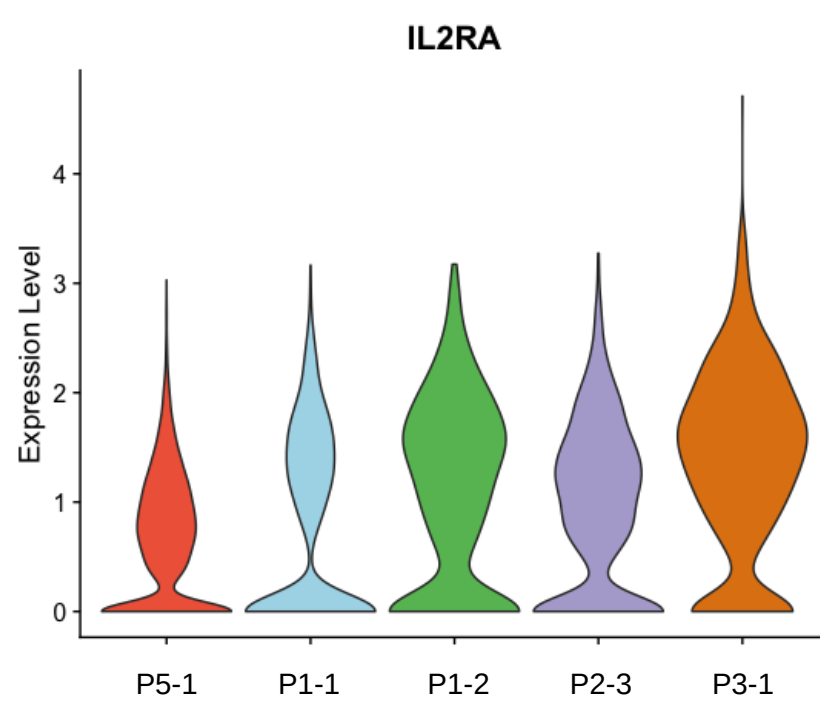
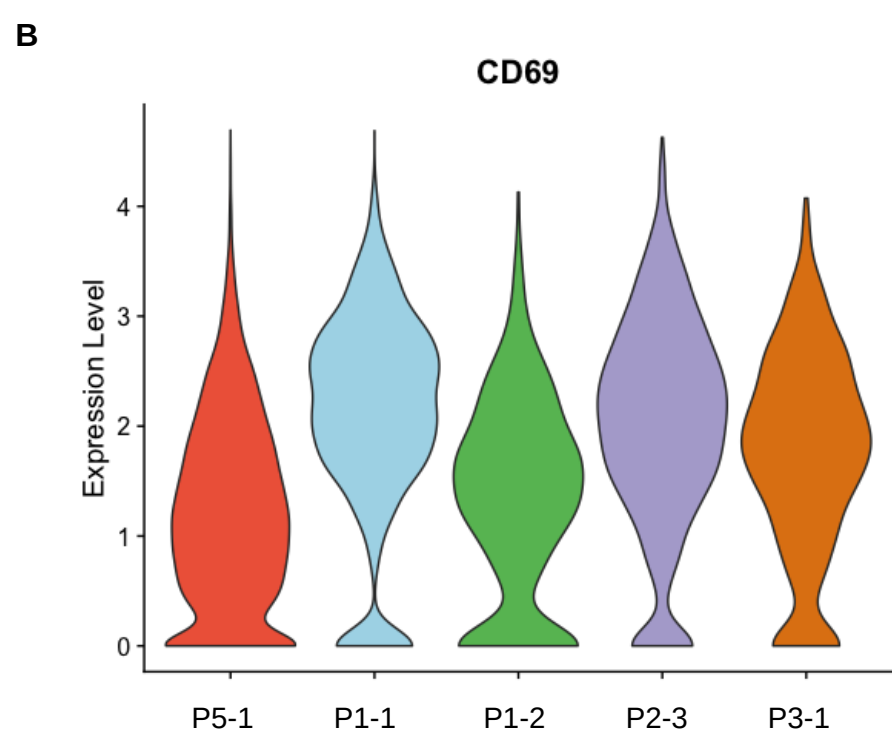
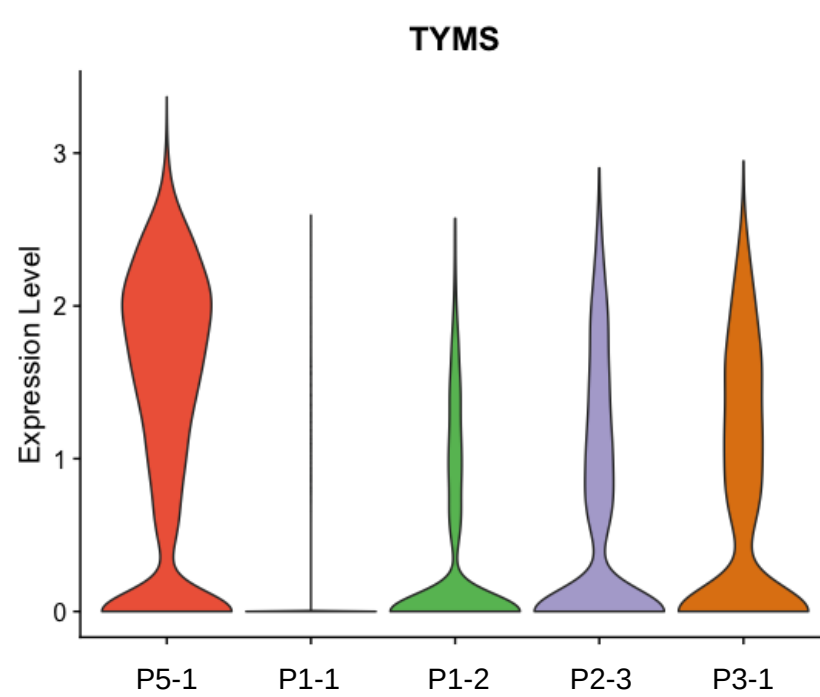
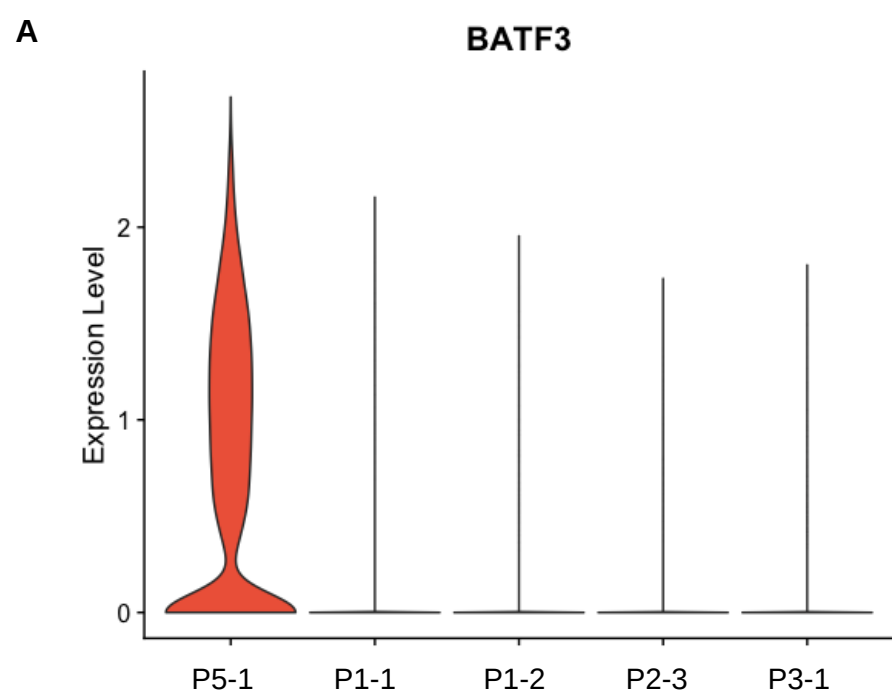
A



B



Supplemental Figure 5. GD2-CAR.15 NKTs derived from healthy donors only underwent a proliferation burst when cryopreserved at an early timepoint after secondary stimulation. A) GD2-CAR.15 NKTs were generated from 12 healthy donors, secondary-stimulated with either A4-4 aAPCs (blue lines) or PBMCs (green lines), then cryopreserved on day 2 post-secondary stimulation (left) or day 10 post-secondary stimulation (right). Thawed products were cultured for 21 days in the absence of exogenous cytokines. Blue arrow at day 7 indicates dilution to 2.5×10^5 cells/well starting concentration when numbers exceeded 3.75×10^5 threshold (dotted line). B) GD2-CAR.15 NKT cell counts at day 7 of assay shown in (A) for cells that were secondary-stimulated with aAPCs or PBMCs and cryopreserved at day 2, day 4, and day 10 after secondary stimulation. Dotted line indicates the threshold for splitting cell culture (3.75×10^5), $p < 0.0001$.



Supplemental Figure 6. GD2-CAR.15 NKTs from patient P5-1 exhibit a memory-like gene expression profile and later stage of activation. Single-cell RNA sequencing analysis and gene expression of pre-infusion cells from patient P5-1 compared to products from four other patients that underwent secondary stimulation with PBMCs (P1-1, P1-2, P2-3, P3-1). A) BAFT3 and TYMS expression. B) CD69 and IL2RA expression. C) HLA-DRA and HLA-DRB5 expression.